

Mingze Li Leukos

Currently publishing under the academic surname “Leukos”. Previously published under “Mingze Li” (Li, M.)

Attention | Bio-inspired Neural Networks | Machine Learning

mingze.li@nyu.edu

Education

Ph.D. in Cognition & Perception

Sept. 2024 – present

New York University, New York, USA

Advisor: Dr. Grace W. Lindsay

Computational and Cognitive Neuroscience Summer School

Jul. 2023

Cold Spring Harbor Asia, Suzhou

- Topics: Neuron and Network Dynamics, Dynamical Systems, Reinforcement Learning, Recurrent Neural Networks, Large Scale Models, Stable Diffusion Models, Decision-making, Attention, Learning and Memory, Motor Control, etc.
- Individual Final Project: *How does Attention Affect Network Dynamics?* [Link to Presentation](#) / [Link to Codebase](#)

Honours B.A. & Sc. in Cognitive Science

Sep. 2020 – May 2023

McGill University, Montreal, Canada

Minor in Mathematics | Concentration in Computer Science

- Relevant Quantitative Coursework: Reinforcement Learning, Applied Machine Learning, Non-linear Dynamics and Chaos, Algorithms and Data Structures, Linear Algebra, Statistics, Multivariable Calculus, Ordinary Differential Equations.
- Relevant Neuroscience Coursework: Human Cognition and the Brain, Introduction to Neuroscience 2.

Neuromatch Academy

Jul. 2022

Remote

Neuromatch Computational Neuroscience Interactive Track Trainee

- Topics: Machine Learning, Dynamical Systems, Modeling, Stochastic Processes, etc.

Research Experiences

Computational Modeling Project: Curiosity-Driven Exploration & Attention in Reinforcement Learning

May 2026 – present

Center for Neural Science, New York University

Advisor: Dr. Grace W. Lindsay

Project: Attention, World-Model Learning, and Curiosity-Driven Exploration in Reinforcement Learning Agents

- Formalize attention as a context-dependent observation-to-state encoder ($s_t = f(o_t; c_t)$), bridging cognitive theories of attention with state representations in reinforcement learning agents
- Design and implement a novel experimental paradigm (Color Cluster Escape Task) in PsychoPy with integrated eye-tracking (PyLink) to study how attention drives hypothesis generation and world-model learning under uncertainty
- Deploy a theory-based reinforcement learning codebase (EMPA) on the NYU Greene HPC cluster via Apptainer containerization; build a custom RL gridworld environment (Quantum Rainbow) to study multidimensional curiosity-driven exploration
- Collaborate across NYU cognitive science labs (Dr. Marcelo Mattar, Dr. Mark Ho) to extend the paradigm toward attention-guided planning and developmental questions

Computational Modeling Project:

Sep. 2024 – present

Reward-Prediction-Error-Guided Attention

Center for Neural Science, New York University

Advisor: Dr. Grace W. Lindsay

Project: Attention-Modulated Reinforcement Learning Model for Feature-Based Selective Attention

- Develop a reinforcement learning model with dynamic attention modulation based on reward prediction error to explain learning trajectories in selective attention tasks
- Design a perceptual front-end with color-tuned filters and implement both single- and multi-focus attention mechanisms, modulated by task volatility and outcome uncertainty
- Fit model behavior to empirical monkey data and analyze RPE-dependent attentional shifts

Technical Consultant

Sep. 2023 – Jan. 2024

SilicoLabs, Toronto, Canada

Advisor: Dr. Blake Richards

Project: 3D Unity-Integrated Platform for Reinforcement Learning and VR Training of Humans and Animals

- Build and train reinforcement learning models in a Unity-supported three-dimensional task environment
- Lead the redesign of a cross-platform Unity-supported reinforcement learning framework, enabling AI model training and testing in 3D simulations of real-world environments, with support for VR-enabled human and animal training

Research Assistant

Sep. 2022 – Jan. 2024

Mila – Quebec AI Institute, Montreal, Canada

Advisor: Dr. Blake Richards

Project: Connectome-to-Model Toolbox: Bridging Brain Architecture with Artificial Networks

- Build and train multi-modal, multi-input neural networks with top-down feedback to model the ventral visual pathway
- Develop a connectome-to-model toolbox in PyTorch that transforms any arbitrarily connected graph into an artificial neural network with brain-like top-down and recurrent connections

Research Assistant

May 2022 – Dec. 2023

Montreal Neurological Institute, Montreal, Canada

Advisors: Dr. Michael Petrides and Dr. Kep Kee Loh

Project: Comparative Analysis of Sulcal Patterns in the Lateral Occipital Cortex across Humans and Chimpanzees

- Label vision-related sulci in fMRI brain data of various subjects from different primate species using FreeSurfer
- Develop data analysis scripts in Python to study sulci similarities and differences across various primate species
- Mount histological primate brain slices

Student Researcher [Honours Thesis Project]

Sep. 2021 – Aug. 2023

Montreal Neurological Institute, Montreal, Canada

Advisor: Dr. Bratislav Misic

Project: Animal Connectome Informed Reservoir Computing of Multiple Tasks

- Build connectome-informed neural networks on 5 different animal connectomes (*C. elegans*, drosophila, mouse, marmoset, macaque) with reservoir computing using Python
- Train the connectome-informed neural networks with 15 cognitive tasks from Neurogym and Gym
- Co-develop the Conn2res Toolbox to train connectome-informed reservoirs on different cognitive tasks in neuroscience

Research Assistant

Sep. 2022 – Apr. 2023

Institut des Neurosciences de la Timone, Marseille, France (remote)

Advisors: Dr. David Meunier and Dr. Kep Kee Loh

Project: Macapype: Python-Based Pipelines for Multi-Modal Brain MRI Anatomical Data Analysis

- Manually quality-check segmentations of marmoset fMRI images generated through the Macapype pipeline
- Correct T1 gray and white pipeline-generated marmoset fMRI images based on actual fMRI images

Research Intern

May 2021 – Sep. 2021

IDG/McGovern Institute at Peking University, Beijing, China

Advisor: Dr. Zhou Yang

Project 1: Decision Making with Brain-Inspired Recurrent Neural Networks

Project 2: Human Psychophysical Experiments on Decision Making

- Co-programmed a bio-inspired RNN model to study decision-making mechanisms in the brain regions of LIP and FEF
- Independently designed and conducted human psychophysical decision-making experiments using eye trackers with Psychtoolbox

Research Intern

Jun. 2017 – Aug. 2017

Oregon State University, Corvallis, USA

Advisor: Dr. Eugene Zhang

Project: C++ Implementation for Illustrating Smooth Surfaces (by Hertzmann, Zorin)

Publications

1. **Leukos, M.L.**, Liang, A., and Lindsay, G.W. Modulation of feature attention by reward prediction error explains value learning behavior. *Journal of Neuroscience* (in revision). Preprint: bioRxiv 2026.04.10.717847.
2. Suárez, L.E., Mihalik, A., Milisav, F., Marshall, K., **Li, M.**, Vértes, P.E., Lajoie, G., and Misić, B. Connectome-based reservoir computing with the conn2res toolbox. *Nature Communications*, 15, 656 (2024). DOI: [10.1038/s41467-024-44900-4](https://doi.org/10.1038/s41467-024-44900-4)
3. Tugsbayar, M., **Li, M.**, Muller, E., and Richards, B. Top-down feedback matters: Functional impact of brain-like connectivity motifs on audiovisual integration. *eLife* (Reviewed Preprint). DOI: [10.7554/eLife.105953](https://doi.org/10.7554/eLife.105953)

Conference Presentations

1. **Leukos, M.L.**, Lindsay, G.W. “Reward-Prediction-Error-Guided Attention Explains Behavioral Learning Curves,” *8th Annual Conference on Cognitive Computational Neuroscience (CCN)*, August 2025, Amsterdam, Netherlands. [accepted as poster and talk]
2. Tugsbayar, M., **Li, M.**, Muller, E., and Richards, B. “Connectome to model: brain-like connections for audiovisual processing,” *Helmholtz International Brain Analytics Laboratory (HIBALL) 7th Annual Workshop*, October 2023, Reykjavik, Iceland.
3. Tugsbayar, M., **Li, M.**, Muller, E., and Richards, B. “Connectome to model: new tool for exploring top-down recurrent model of human vision,” *Healthy Brains Healthy Lives Symposium*, May 2023, Montreal, Canada.
4. Suárez, L.E., **Li, M.**, Marshall, K., Mihalik, A., Vértes, P., Lajoie, G., and Mišić, B. “Conn2res: A toolbox for Connectome-based Reservoir Computing,” *Organization of Human Brain Mapping (OHBM)*, June 2022, Glasgow, Scotland, UK.
5. **Li, M.**, Suarez, L.E., and Misić, B. “Animal Connectome Informed Reservoir Computing of Multiple Tasks,” *UNIQUE Student Symposium*, May 2022. [accepted]
6. **Li, M.**, Suarez, L.E., and Misić, B. “Animal Connectome Informed Reservoir Computing of Multiple Tasks,” *National Integrative Research Conference*, April 2022. [first place presentation award]. [Link to Poster](#).
7. **Li, M.**, Suarez, L.E., and Misić, B. “Animal Connectome Informed Reservoir Computing of Multiple Tasks,” *Undergraduate Poster Showcase*, March 2022.
8. **Li, M.**, Suarez, L.E., and Misić, B. “Animal Connectome Informed Reservoir Computing of Multiple Tasks,” [Webinar], *McGill Chinese Psychology Student Association*, Montreal, QC. (invited talk)

Professional Service

Local Organizing Committee

2026

Cognitive Computational Neuroscience (CCN) 2026

- Member of the local organizing committee for the 9th Annual Conference on Cognitive Computational Neuroscience (CCN).

Organizer

2026

NYU Psychology Master's Thesis Event (with Dr. Grace W. Lindsay)

- Co-organized the NYU Psychology master's thesis event, coordinating logistics and presentation sessions.

Leadership Experiences

Co-Editor-in-Chief

Sep. 2022 – Sep. 2023

Cognitive Circuits: McGill University, Montreal, Quebec

Editor Position, Sep. 2020 – Sep. 2022

- Oversee all operations of *Cognitive Circuits*, the McGill undergraduate peer-reviewed cognitive science academic journal
- Organize a cognitive science research conference to showcase undergraduate research for 60 students
- Review all article submissions and assist editors with outreach and contacting professors to peer-review articles
- Promote research accessibility to all McGill students (bi-weekly research Q&A channel, podcast interviews with graduate students, lab spotlights on social media)

Youth Programming Program Founder & Teacher

Sep. 2018 – Jun. 2019

Corvallis, Oregon

- Design and teach free weekly coding lessons (Scratch and C++) for 40 elementary school students
- Promote gender and racial equity in youth STEM education by initiating a free coding program in local elementary schools

Honours and Awards

2025 TPCN (Theoretical Principles in Cognition and Neuroscience) Fellowship

2024 Henry M. MacCracken Fellowship for Doctoral Students at NYU

2023 Dean's Multidisciplinary Undergraduate Research List

2023 First Class Honours in Cognitive Science

2022 Emily Ross Crawford Scholarship \$1,000

2022 Student Research Initiative Scholarship \$1,000

2022 Krav Maga Practitioner Level One Certificate

2022 First Place Presentation Award at National Integrative Research Conference \$100

2021 Faculty of Science Scholarship \$300

2020	Hugh Brock Scholarship	\$20,000
2020	McGill University Entrance Scholarship	\$3,000
2019	National Scholastic Writing Competition, Honourable Mention Award	
2019	Member of Cum Laude Society	
2018	First Place in Data Structures and Algorithms	
2018	First Place in Honours Latin One	

Skills

Programming: Python (PyTorch), Java, C++, C, C#, Bash, MATLAB (Psychtoolbox), JavaScript, HTML, Unity

Data Processing: FreeSurfer, ITK-SNAP

Wet Lab: Primate fMRI data labeling (human, macaque, chimpanzee, marmoset, etc.), psychophysical experiments using eye trackers, histological processing

Natural Languages: English (fluent), Mandarin (fluent), French (basic)

Interests: Krav Maga, collecting neuro-related items ([Link to Collection Gallery](#)), philosophy, story-writing, reading, drawing, skiing, camping, biking, percussion.